SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I G.Naveen Kumar Reddy(192525288) (Name / Reg. No.) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: I G.Naveen Kumar Reddy ____

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks:

1. Define Cloud Computing and Explain four characteristics.

Definition:
Cloud Computing means Using Computing devices like Storage, Services and Software through internet instead of owning a physical Computer and Server.

Four Characteristics

1. On-Demand Self-Service.
Users Can access cloud resources whenever they need them.

2. Broad Network Access
Cloud Services Can be accessed over the internet Using laptops, Smartphones and other devices.

3. Resource Pooling:
A cloud Provider Shares Computing resources among multiple Users while keeping their data logically Separated.

4. Rapid elasticity
Resources Can be increased or decreased according to demand.

2. Trace the evolution of Cloud Computing from hardware Evolution to internet Software evolution, and Show how these changes Enabled modern cloud platforms.

A. Cloud Computing developed through these Stages.

1. Main frame Computers :.
One large Computer Served Many Users.

2. Personal Computers :.
People Started Using individual Computers.

3. Client-Server Computing
Computers accessed Services from Central Servers.

4. Virtualization : One physical Server Could run Multiple virtual Machines.

5. Internet and web applications :.
Users Could access Services Online.

6. Modern Cloud Computing :.
Companies provide Servers, Storage, and Software through the internet.

3. Explain the role of Server Virtualization in cloud Computing. Differentiate between Virtualization as an Enabling technology and Cloud Computing as a Service Model.

A. **Server Virtualization**

It is a technology that divides One physical Server into multiple Virtual Machines.

Role of Virtualization.

1. User hardware efficiently.
2. Reduces the number of physical Servers needed.
3. Saves Cost.
4. Allows Multiple Operating Systems to run on one Server.

Difference

Virtualization	Cloud Computing
• Creates Virtual Machines.	• Provides Computing Services over a Network.
• It is a technology.	• It is a Service delivery model.
Ex:- VMware.	• Ex:- AWS.

4. Discuss the role of data Center in delivering cloud Services.

A. A data Center is a place Containing Servers, Storage devices, and Networking Equipment Used to provide Computing Services.

Roles of data Centers

1. Storage :- Stores files and databases.
2. Processing :- Processes data and runs applications.
3. Connectivity :- Connects cloud services to users through the internet.
4. Security :- Protects systems and data.
5. Reliability :- Uses backup systems to reduce service interruptions.
6. Scalability :- Adds Resources when demand increases.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world Example for each model.

A.

Model	Full form	Simple Meaning
IaaS	Infrastructure as a Service.	Provides Virtual Servers and storage.
PaaS	Platform as a Service	Provides a platform to develop applications
SaaS	Software as a Service	Provides ready-to-Use Software Online.
XaaS	Anything as a Service.	Provides different IT Services over a Network.

Explanation :

1. IaaS :- Users rent Servers and Storage.
2. PaaS :- Developers Use a platform to build applications.
3. SaaS : Users directly Use Software through the internet.
4. XaaS : Different IT resources are Provided as Services.

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